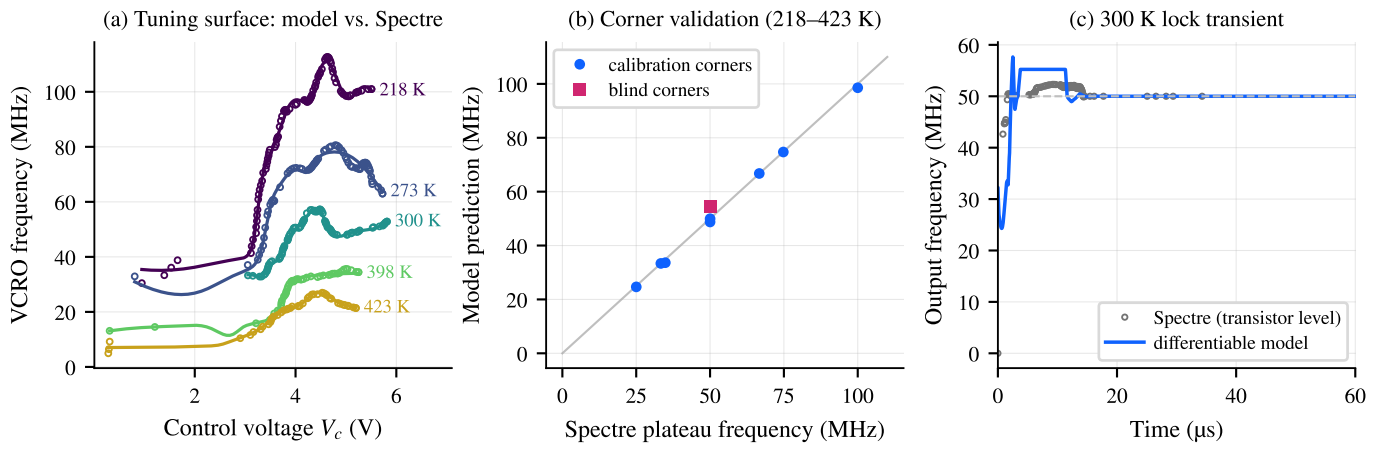
(a) Tuning surface: model vs. Spectre
VCRO frequency (MHz)
Control voltage $V_c$ (V)
218 K
273 K
300 K
398 K
423 K
(b) Corner validation (218–423 K)
Model prediction (MHz)
Spectre plateau frequency (MHz)
calibration corners
blind corners
(c) 300 K lock transient
Output frequency (MHz)
Time (µs)
Spectre (transistor level)
differentiable model